Mohammad Ali Safaei

CONTACT

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EDUCATION

University of Tehran – Tehran, Iran *(Ranked #368 in QS Rankings)* ***2021 – 2024***
M.Sc. in Mechanical Engineering |Applied Mechanics specialization

Thesis Title: Distinctive Electromechanical Performance of Metamaterial Dielectric Elastomer Actuators with TPMS Geometries (Link).
Supervisor(s): Dr. Mostafa Baghani, Dr. Majid Baniassadi.
GPA: 18.15/20.00. (Converted: 3.88/4.00) (Ranked 7th among more than 40 students)

Bu-Ali Sina University – Hamedan, Iran ***2016 - 2021***
B.Sc. in Mechanical Engineering

Thesis Title: Optimization of Energy Absorption in Empty and Foam-Filled Circular Thin-Walled Sandwich Structures (Link).
Supervisor(s): Prof. Ali AlaviNia.
GPA: 15.66/20.00. (Converted: 3.12/4.00). (Last 2 Semesters GPA: 3.51/4.00)

RESEARCH INTERESTS

- Finite Element Analysis, Finite Difference Method
- Constitutive Modelling of Non-linear Solids
- Micromechanics, Multi-scale Modelling
- Metamaterials, Architected Materials
- Machine Learning , Deep Neural Networks
- Additive Manufacturing, 3D bioprinting

SKILLS

Programming: Python, MATLAB, Fortran, C++, MAPLE.
CAD software (Link): SOLIDWORKS ,CATIA.
CAE software (Link): ABAQUS, COMSOL Multiphysics, , LS-DYNA.
3D printing: Simplify3D, Cura.
Languages: *Persian* (Native), *English*: IELTS Academic, Overall Band 7.5 (L: 7.0, R: 9.0, W: 7.0, S: 6.5).
Documentation Software: MS Office (Word, PPT, Excel).
Hands-on Experiences: Casting, Welding, Machining, Turning, Milling, Nondestructive testing (NDT).

PUBLICATIONS

Research Article:

In Silico Actuation Performance Investigation of Dielectric Elastomers with TPMS Geometries (Link)

Optimization of Electromechanical Performance in TPMS-Based Dielectric Elastomers Using Genetic Algorithm (*In Completion*)

Book Chapter:

Towards Responsive Polymer Actuators Based on Novel Composite Structures (*Submitted*)

EXPERIENCES

School of Mechanical Engineering, University of Tehran, Tehran, Iran ***2022 – Now***
|*Research Assistant - SMS lab*|
Supervised and mentored students in their projects and oversaw junior students in part-time 3D printing operations.
Authored Section 1.3 of the book "*Computational Modeling of Intelligent Soft Matter*" (Link).

|*Teaching Assistant*|
FEM (Link), Continuum, Non-linear Solid Mechanics (Link) and Research Methodology (Link) Courses.

Delivered lectures and presentations, designed and assessed projects and exams, and provided feedback on students' "proposal writing" assignments.

Voluntary Job ***2023 – Now***

|Assistant *Reviewer*|

Reviewed over 20 submissions for journals such as *Chemical Engineering Journal*, *Additive Manufacturing*, *Advanced Engineering Materials*, and *Polymers*.

Emamat High School ***2024 – Now***

|*General English Teacher (part-time)*|

IKCO Company – Tehran, Iran ***2021***

|*Intern*|

A brief review of my internship: (Link)

MDFICO (Knowledge-Based Enterprise Company - Grade A) – Tehran, Iran ***2018 – 2020***

|*Junior Engineer*|

Provided services at 1- Casting Shop: Low-pressure and gravity casting of magnesium and aluminum alloys. 2- QC Department: DT & NDT test.

HONORS & AWARDS

Top 1%, out of more than 10,000 applicants in the M.Sc. Nation-wide University Entrance Exam (**Full scholarship**).
Top 5%, out of more than 190,000 applicants in the B.Sc. Nation-wide University Entrance Exam (**Full scholarship**).
Awarded within the top state candidates in the National Kharazmi Elite Youth Festival. (Poetry-Literature)
Performed as a Santur player in the talented students orchestra during elementary and secondary school.

LICENCES & CERTIFICATES (Link)

ICDL - Python: Fundamental & Advanced - COMSOL: Multiphysics - MATLAB: fundamental - Piping principles and design - Pneumatic Technician - ISO 9001:2015 Internal Auditor

ACADEMIC PROJECTS

ML Algorithms & Neural Networks Implementation

Artificial Intelligence Course – Course Project

- Developed ML Algorithms (Link), MLP and CNN algorithms Neural Networks (Link), RL algorithms like DQNs (Link).

Investigating Linear Viscoelastic Behavior of TPMS Structures.

Viscoelasticity Course – Final Course Project

- Evaluated the visco-hyperelastic response of complex geometries under thermomechanical loading conditions (Link).

Transient swelling of pH-sensitive Hydrogels – Shape Memory Polymers under Thermo-Mechanical Loading.

Non-linear Solid Mechanics Course – Final Project & Course Projects

- Developed and implemented semi-analytical models for transient swelling of pH-sensitive hydrogels in COMSOL.
- Designed and implemented monolayer SMPs with various actuation modes under thermomechanical loading in COMSOL (Link).

FEM Implementation in MATLAB, Python, ABAQUS (Link).

FEM: Intro & Advanced Course – Course Projects

- Developed various UEL & UMAT subroutines for soft materials.
- Implemented Finite Element Algorithms in MATLAB & Python from scratch.

REFERENCES

Dr. Mostafa Baghani. Professor, University of Tehran, Iran.
Scopus, E-mail.

Dr. Majid Baniassadi. Professor, University of Tehran, Iran.
Scopus, E-mail.

Dr. Mahdi Bodaghi. Associate Professor, School of Science & Technology, Nottingham Trent University, UK.
Scopus, E-mail.